

E7 Guest System — Flow Chart

Self-contained rendered version: [E7GuestSystem-FlowChart.html](#) Mermaid source below for version control. Render at [mermaid.live](#).

1 · Visitor Check-In Flow (Web / Public)

```

flowchart TD
    A([Visitor arrives at location]) --> B{Has QR code  
or shared link?}
    B -- Scans QR  
or opens link --> C[Land on  
visitor form page]
    C --> D[Fill out form  
• Name(s)  
• Party size  
• Local / Moving  
• Friends/Family in church  
• Reason for visit  
• Prayer request / Follow-up opt-in]
    D --> E{Opted in to  
SMS follow-up?}
    E -- Yes --> F[System sends automated  
SMS welcome message  
via Twilio  
using admin template]
    E -- No --> G[Record visitor only  
No SMS sent]
    F --> H[Log visitor registration  
in database  
• Status: new  
• Timestamp  
• IP + consent record]
    G --> H
    H --> I[Push notification sent  
to account admin / staff]
    I --> J[Visitor lands on  
Thank You screen]
    J --> K([End])

    style C fill:#0d4429,stroke:#2e7d32,color:#e6edf3
    style F fill:#0c2d57,stroke:#1565c0,color:#e6edf3
    style G fill:#4e3b00,stroke:#f9a825,color:#e6edf3
    style H fill:#3b1a52,stroke:#7b1fa2,color:#e6edf3
    style I fill:#0c2d57,stroke:#1565c0,color:#e6edf3
    style J fill:#0d4429,stroke:#2e7d32,color:#e6edf3
  
```

2 · Account Admin Onboarding Flow

flowchart TD

A([Admin signs up]) --> B{Auth method}

B -- Email + Password --> C[Convex email/password auth]

B -- Google OAuth --> D[Convex Google provider]

B -- Apple OAuth --> E[Convex Apple custom provider]

C --> F[Account created]

- Account ID
- Stripe customer
- Default: Starter tier]

D --> F

E --> F

F --> G[Admin sets up organization]

- Church or business name
- Timezone
- Notification preferences]

G --> H[Create Location]

H --> I[Configure form fields]

- Standard templates
- Enable/disable fields
- Set required or optional]
- I --> J[Set admin SMS reply template]
- Max 140 chars
- Variables: visitor name, church name, etc.]

J --> K[Generate QR code]

- Signed short-lived URL
- Download PNG / print-ready PDF]

K --> L[Share link or print QR
for visitor check-in]

L --> M([Location live and
ready for visitors])

style F fill:#0c2d57,stroke:#1565c0,color:#e6edf3

style H fill:#0d4429,stroke:#2e7d32,color:#e6edf3

style I fill:#4e3b00,stroke:#ef6c00,color:#e6edf3

style J fill:#4e3b00,stroke:#ef6c00,color:#e6edf3

style K fill:#4e3b00,stroke:#ef6c00,color:#e6edf3

style L fill:#0d4429,stroke:#2e7d32,color:#e6edf3

style M fill:#1a3a1f,stroke:#388e3c,color:#e6edf3

3 · Staff Follow-Up Flow (App / Web Dashboard)

```

flowchart TD
    subgraph Trigger
        A[Push notification  
New visitor registered] --> B[Staff opens  
app or web dashboard]
    end

    B --> C{Action}
    C --> D[View visitor list
    • Activity log
    • Filter by date / location / status]
    D --> E[View visitor profile
    • Full form responses
    • Notes from other staff]
    E --> F[Start message thread
    • SMS to visitor via Twilio
    • In-app chat if visitor responds]

    F --> G([Review complete])
    G --> H[Add notes / update  
contact info and status]
    H --> I{Visitor responds?}
    I -- Yes --> J[Continue conversation  
in message thread]
    I -- No --> K[Mark as contacted  
Set follow-up reminder]

    K --> L{Set reminder?}
    L -- Yes --> M[Schedule reminder
    • Per visitor or general
    • Push / email / SMS channel]
    L -- No --> G

    M --> N(Reminder fires at  
scheduled time)
    N --> O[Staff sees reminder  
in dashboard / app]
    O --> P{Take action?}
    P -- Yes --> F
    P -- No --> Q[Dismiss / mark complete]

    Q --> R([End])

    style A fill:#0c2d57,stroke:#1565c0,color:#e6edf3
    style B fill:#0d3320,stroke:#388e3c,color:#e6edf3
    style C fill:#4e3b00,stroke:#f9a825,color:#e6edf3
    style F fill:#0d3320,stroke:#388e3c,color:#e6edf3
    style H fill:#3b1a52,stroke:#7b1fa2,color:#e6edf3
    style M fill:#0c3440,stroke:#00838f,color:#e6edf3
    style R fill:#3d4148,stroke:#455a64,color:#e6edf3

```

4 · Inbound SMS Processing Flow

```

flowchart TD
    A([Visitor sends  
inbound SMS]) --> B{Twilio webhook  
receives message}
    B --> C[Parse phone number  
Match to visitor record]
    C --> D{Message content?}

    D -- Opt-out keyword STOP  
UNSUBSCRIBE or QUIT --> E[Update visitor status  
Set to opted_out  
Log opt-out event]
    E --> F[Send final acknowledgement SMS  
You have been unsubscribed]

    D -- Normal message --> G[Create or append to  
message thread  
visitor to staff]
    G --> H[Push notification  
to assigned staff]
    H --> I[Staff replies from  
app or web dashboard]
    I --> J[Outbound SMS sent  
via Twilio]

    D -- Unrecognized keyword --> K[Log as unknown  
Optional: auto-reply  
with help text]

    F --> L([End])
    J --> L
    K --> L

    style A fill:#0c2d57,stroke:#1565c0,color:#e6edf3
    style E fill:#3b0a0a,stroke:#c62828,color:#e6edf3
    style F fill:#3b0a0a,stroke:#c62828,color:#e6edf3
    style G fill:#0d3320,stroke:#388e3c,color:#e6edf3
    style H fill:#0c2d57,stroke:#1565c0,color:#e6edf3
    style J fill:#0c2d57,stroke:#1565c0,color:#e6edf3
    style K fill:#4e3b00,stroke:#ef6c00,color:#e6edf3

```

5 · System Architecture Overview

```

flowchart LR
    subgraph Frontend
        F1[Vite plus React PWA  
Visitor forms and admin dashboard]
    end

```

```

    F2[Expo App
Staff mobile CRM and messaging]
end

```

```

    subgraph Backend
      B1[Convex
Auth, DB, Server functions]
      B2[Twilio
SMS send, receive, webhooks]
      B3[Resend
Transactional email]
      B4[Stripe
Subscription billing]
    end

```

```

    subgraph Infrastructure
      I1[Netlify
Frontend hosting and CI/CD]
      I2[GitHub Actions
Lint, test, deploy]
      I3[Sentry
Error tracking]
      I4[PostHog
Analytics]
    end

```

```

F1 -- API calls --> B1
F2 -- API calls --> B1
B1 -- SMS send --> B2
B2 -- Webhooks --> B1
B1 -- Email send --> B3
B1 -- Billing --> B4
B1 -- Deploy --> I1
B1 -- CI/CD --> I2
B1 -- Error logs --> I3
F1 and F2 -- Usage data --> I4

```

```

style F1 fill:#0d3320,stroke:#27ae60,color:#e6edf3
style F2 fill:#0d3320,stroke:#27ae60,color:#e6edf3
style B1 fill:#0c2d57,stroke:#2980b9,color:#e6edf3
style B2 fill:#4e3b00,stroke:#e67e22,color:#e6edf3
style B3 fill:#4e3b00,stroke:#e67e22,color:#e6edf3
style B4 fill:#4e3b00,stroke:#e67e22,color:#e6edf3
style I1 fill:#1a1a40,stroke:#8e44ad,color:#e6edf3
style I2 fill:#1a1a40,stroke:#8e44ad,color:#e6edf3
style I3 fill:#1a1a40,stroke:#8e44ad,color:#e6edf3
style I4 fill:#1a1a40,stroke:#8e44ad,color:#e6edf3

```